Readable Depth Without the Calibration Map: Formula Placement on Heavy-Hex Devices, Measured by a Circuit's Own Conserved Charge

TN Holdings
love.os.architect@proton.me
https://github.com/TN-Holdings-LLC/psf-zero

Draft 0.3 — 31 October 2026 (not for release)

Abstract

Compilers for today's quantum processors choose a layout by reading the day's calibration: error rates per qubit and per coupler. A classical computer offers no such map, and a map is stale as soon as it is read. We ask how deep a circuit stays readable when no placement reads a quality number. We place a ring of CZ gates on IBM's heavy-hex devices in three ways that see only the coupling graph, the basis and which elements have failed: a formula that puts the ring on a lattice cycle (the *rule*), the ring folded onto the same qubits (*zipper*), and Qiskit's level-2 search given the coupling map without error rates (*blind Qiskit*). As the workload we use the kicked Ising ring at $J = \pi/4$, whose exact answer is known at any size, and as the probe a weight-2 charge conserved by the circuit itself: its decay, read with a four-input combination that cancels readout and first-order relaxation, gives the per-step damping without any calibration data. In declared tests on `ibm_marrakesh` and `ibm_kingston` (29 October 2026; $n = 20$ and 36; $L = 24$, 50, 100 steps, up to 3600 CZ per circuit) the rule kept more of the charge than both other placements in all 24 comparisons. Its readable depth, the $1/e$ decay length L^*, was 56–57 steps on marrakesh and 81–90 on kingston, against 15–26 for the others: 30–73 steps deeper. In an earlier test with a 16-layer classifier at $n = 13$–24 the rule beat blind Qiskit at 11 of 12 sizes and tied one. The noise model we declared did not hold: one depolarizing rate did not describe the decay on either device, and the zipper's penalty, 3.5–3.7 times the rule's decay rate on both devices, was larger than its gate count and duration predict (2.2–2.4). A second day (30 October) added a mirror echo of the same ring with a longitudinal field that breaks integrability, sizes up to $n = 72$ and the global parity. The rule's lead survived: 71 wins, 11 ties and no loss in 144 comparisons (62 not judged, both arms at the noise floor). The penalty fixed after the first day did not carry over to the echo (zipper ratio 1.7–2.7 against 3.3–3.9), and the declared parity volumes were too optimistic. On `ibm_kingston` the rule's $n = 40$ ring carried a run of 18 sites whose echo was near random or inverted, the same 18 sites in three runs of the byte-identical circuit over three hours, while the calibration showed nothing there and the same qubits were healthy on the $n = 20$ ring. Gate twirling, which leaves the ideal circuit unchanged, turned the inverted part random: the cause is on the device, not in the circuit. These sites set the readable depth at $n \geq 40$ (13–16 steps against 22–26 at $n = 20$, 36) and erased the parity at $n = 40$. A map read from the calibration could not have avoided them.

1 Introduction

- Map-aware placement is the default (Qiskit's VF2 post-layout and noise-aware routing read error rates). Its gains are measured against the same map that chose the layout; the selection is optimistic by construction (the optimizer's curse [3]), and the map ages.

- The question here: without any quality number, how deep can a circuit be read, and does the placement change that depth? We call a placement *map-free* if it reads only the coupling graph, the basis and the failed elements. Map-aware placement is a special condition and is always labelled so.
- What is new: (i) a map-free comparison of placements by a calibration-free decay measurement on the circuit itself; (ii) the readable depth L^* and the gain ΔL as the reported quantity; (iii) declared predictions, with the failed ones reported. What is not new: reading noise from conserved quantities [5, 6], layer fidelity and size-independent layer errors [7], the noisy free-fermion solution [8], and the $J = \pi/4$ kicked Ising circuit on IBM devices [11].
- A picture from underwater acoustics guides the reading (Section 2.5). Sound in the ocean fades with distance at a rate set by its frequency and by the water; low tones in the deep sound channel carry for thousands of kilometres. Here the depth of a circuit plays the distance, the Pauli weight of what is read plays the frequency, the placement plays the water, and a conserved charge plays the sound channel. It is an analogy for the reader, not a physical identity.
- Companion work: the formula schedules and their gate counts [1].
- Second day (30 October): a non-integrable mirror echo, sizes to $n = 72$, the global parity, and a defect that no calibration number shows (Sections 7–9).

2 Setting

2.1 Workload

The kicked Ising ring, one step $S = (R_X(2h)\, Z)^{\otimes n}\, \mathrm{CZ}_{\mathrm{ring}}$, $J = \pi/4$, $h = 0.6$. It is a free-fermion circuit; its outputs are computed exactly at any n by a recursion, which also gives the outputs under uniform depolarizing noise exactly [8]. [TODO: one paragraph with the recursion, from toqb.kising]

2.2 Placements (no quality number read)

- **Rule** (`fold:rule`): the ring on a cycle of the heavy-hex lattice, built by formula from the lattice's cells [1]; n CZ per layer.
- **Zipper** (`fold:zipcyc`): the ring folded onto the same cycle's qubits; $5n/2 - 4$ CZ per layer.
- **Blind Qiskit** (`qiskit:2:blind:template`): Qiskit level 2, `seed_transpiler`=0, given the working couplers in both directions and the basis only.
- Failed elements (error = 1) are removed for every placement; nothing else of the calibration is used.

2.3 Probe P1: the circuit's own conserved charge

$$Q = \sum_j \big(Z_j Z_{j+1} + b\, Y_j Z_{j+1} + c\, Z_j Y_{j+1}\big), \qquad b = c = -1.28608 \ (h = 0.6),$$

the weight-2 conserved charge with the largest ZZ share (a noiseless criterion; the same coefficients for $n = 6$–40). Inputs $|0\cdots0\rangle$, $|1\cdots1\rangle$ and the two Néel states; three measurement settings; every record carries its own readout reference on the physical qubits it measures. The four-input ratio

$$\rho(L) = \frac{[Q(0\cdots0) + Q(1\cdots1) - Q(\text{Néel}) - Q(\overline{\text{Néel}})]_L}{[\,\cdot\,]_{\text{ref}}}$$

cancels readout within a record and the first-order relaxation bias. Under uniform depolarizing noise $\rho(L) = (1 - 4p/3)^{2L}$.

2.4 Readable depth

L^* is the $1/e$ decay length of a one-parameter fit $\rho(L) = e^{-L/L^*}$ over $L = 24, 50, 100$ (weighted least squares on the ρ scale). $\Delta L_X = L^*_{\text{rule}} - L^*_X$ and $\kappa_X = L^*_{\text{rule}}/L^*_X$, the ratio of per-step decay rates. L^* is a property of a weight-2 quantity; wider outputs decay faster (Section 10).

2.5 An acoustic reading of the decay

Under uniform depolarizing noise of rate p per qubit and step, a Pauli string of weight w is multiplied by $(1 - 4p/3)^w$ at every step. The attenuation per step therefore grows with the weight, as the absorption of sound in sea water grows with frequency [9]. Table 1 lists the correspondence we use.

sound in the ocean	a circuit on a device
distance travelled	depth L (steps)
absorption coefficient; attenuation length	per-step decay rate; L^*
frequency (higher tones fade sooner)	Pauli weight of the observable (wider observables fade sooner)
the water (temperature, salinity)	the placement (the zipper absorbs about 3.6 times as much as the rule)
geometric spreading loss	operator spreading: a local output grows in weight before the noise acts on it
the deep sound (SOFAR) channel	a conserved charge: the circuit does not spread it; only the noise damps it
sonar equation and detection range	signal against shot noise; the readable depth at a shot budget

Table 1: The analogy used in this paper. Where it fails: sea-water absorption grows roughly as the square of the frequency, with relaxation peaks, and is an empirical formula; the weight dependence here is linear in the exponent and follows from the noise model.

In the language of the sonar equation [10], a reading at depth L is detected when the signal exceeds twice its shot-noise standard error. With $\rho(L) = e^{-L/L^*}$ and a per-shot spread σ_1 (so that $\text{SE} = \sigma_1/\sqrt{S}$ at S shots),

$$L_{\text{read}}(S) = L^* \ln \frac{\sqrt{S}}{2\sigma_1}.$$

For probe P1, $\sigma_1 \approx 0.2$–0.26, so $L_{\text{read}}(1000) \approx 4L^*$. That lies beyond the depths we measured and is not claimed; we report L^*, which needs no budget and no threshold. The formula does show why the gain matters: ΔL in L^* becomes about four times as many steps of detection range at 1000 shots, and a placement that absorbs κ times as much reaches κ times fewer steps with the same shots.

2.6 Declarations

Every device test was declared (SHA-256 and CET time) before its calibration was fetched; the declarations, records and raw counts are in TOQB [2]. [TODO: table of declarations and job ids]

3 Classifier test without the map (D6)

TOQB's ring classifier, $L = 16$, $n = 13$–24, 64 inputs × 250 shots, `ibm_kingston`, 28 October 2026; no arm read a quality number and no simulator was used. Rule against blind Qiskit: 11 wins, 1 tie (at $n = 23$), no loss. Rule against the zipper on the same qubits (even n): 3 wins, 3 ties, no loss. Blind Qiskit kept no measurable signal at 8 of 12 sizes; its circuits ran 22–108 µs against the rule's 5.6–9.3 µs. For reference, with the map (D5 part A, $n = 20$, a different hour): rule 0.943 ± 0.044, zipper 0.600 ± 0.044, map-aware Qiskit 0.765 ± 0.099 (rule − Qiskit within 2 SE). [TODO: table of D6 from TOQB results/device-d6]

4 The noise wall on two devices

	marrakesh $n = 20$	marrakesh $n = 36$	kingston $n = 20$	kingston $n = 36$
L^* rule	56.5 ± 0.9	56.6 ± 0.7	81.3 ± 1.3	89.7 ± 1.1
L^* zipper	15.1 ± 0.4	15.7 ± 0.3	22.6 ± 0.4	25.6 ± 0.3
L^* blind Qiskit	26.4 ± 0.5	15.7 ± 0.3	22.0 ± 0.4	17.0 ± 0.3
ΔL vs zipper	+41.3	+40.9	+58.7	+64.1
ΔL vs blind Qiskit	+30.1	+40.9	+59.3	+72.7
κ zipper (g, t)	3.73 (2.30, 2.27)	3.60 (2.39, 2.25)	3.60 (2.30, 2.24)	3.50 (2.39, 2.22)
rule wins on ρ	6/6	6/6	6/6	6/6

Table 2: Probe P1 on two devices, 29 October 2026. g, t: the zipper's gate and duration ratios to the rule. 1000 shots per circuit; jobs of 92 and 90 s. [TODO: full ρ tables in an appendix]

- **Benchmark number:** the rule won all 24 charge comparisons; $\Delta L = +30$ to $+73$ steps.
- **Device user's number:** at $n = 36$, $L = 100$ (3600 CZ) the rule's charge was 0.178 ± 0.006 (marrakesh) and 0.314 ± 0.006 (kingston); the zipper's and blind Qiskit's were zero within 2 SE.

5 Declared predictions that failed

- **One depolarizing rate.** Neither device followed $\rho = (1 - 4p/3)^{2L}$ with one p within 2 shot-noise SE: marrakesh against its own p from $L = 24$ (1 of 5 inside), kingston against $p = 0.0051$ from the previous day (1 of 6). The per-segment decay rates scatter by 10–20 % with no common direction, and kingston's rate had improved since the previous day; the declared tolerance (shot noise only, about 1 %) left no room for either.
- **Size independence of the rule.** SAME at all three depths on marrakesh; DIFFERENT at $L = 24$ and 50 on kingston, where the $n = 36$ ring was better than the $n = 20$ ring.
- **The structure bracket.** The zipper's κ was declared to lie between its gate ratio and its duration ratio (2.2–2.4). It was 3.5–3.7 on both devices.

- **Probe P2** ($\langle X_0 \rangle$, 250 shots) was at the noise floor on both devices.

6 The formula fixed after the run

The following changes were made after seeing the results. They are fitted to the same data and are evidence only once they predict a later run; they are declared for the next stage.

1. **Self-reference with a model band.** The rule's rate is read from its own $L = 24$ point; deeper points are predicted with the rate $\pm 20\,\%$ (the observed scatter), plus 2 SE of shot noise.

2. **A constant zipper penalty.** $L^*_{\text{zipper}} = L^*_{\text{rule}}/3.6$, with $\kappa \in [3.3, 3.9]$, in place of the structure bracket.

3. **Map-aware, for reference only (special condition).** $1/L^* = c\times$ the median CZ error of the rule's couplers on the day's calibration.

fixed formula	on its own data	fitted on one device, applied to the other
1. own $L = 24$ point, rate $\pm 20\,\%$	8/8	(no device constant)
1b. $n = 20$ ring's $L = 24$ point to the $n = 36$ ring	6/6	(no device constant)
2. zipper $\kappa = 3.6 \pm 0.3$	4/4	4/4
3. map-aware: $1/L^* = c\times$ median CZ error	fits by construction	1/4 (-27 to $+37\,\%$)

Table 3: The fixed formulas on 29 October 2026 (post-hoc). Formula 3 reads the calibration and failed across devices: the day's error map did not predict the readable depth better than the circuit's own short-depth point.

Formula 2 was then declared for the next day's probes. It failed on both: the zipper's ratio was 1.7–2.7 on the echo (Section 7) and 1.7–2.6 on the parity (Section 9). The penalty measured on the charge is a property of that probe, not of the placement alone.

7 Second day: a non-integrable echo and larger rings

7.1 Probe P3: the mirror echo

The step gets a longitudinal field, $S = (R_X(2h)\, Z\, R_Z(2h_z))^{\otimes n}\, \mathrm{CZ}_{\text{ring}}$, which breaks integrability for $h_z \neq 0$ and leaves the two-qubit structure unchanged (R_Z commutes with the CZ ring). The echo runs L steps forward, flips every qubit ($W = X^{\otimes n}$), and runs L steps of $(WSW)^{-1}$. Since WSW is S with every angle negated,

$$U = (WSW)^{-L}\, W\, S^L = W\, S^{-L}\, W\, W\, S^L = W,$$

so the noiseless output is the flipped input at any n and h_z, and no simulation is needed. The flip also keeps a compiler from cancelling the two halves. Inputs and readout reference are as for P1; the reported quantity is $\rho_E(D)$, the four-input ratio of $\sum_j (-z_j)\langle Z_j \rangle$ to its reference, at full depth $D = 2L = 24$, 50, 100, and L^* is its decay length in the same steps as before. P3 reads weight-1 outputs; it is not P1's charge and its L^* is not comparable with Section 5's.

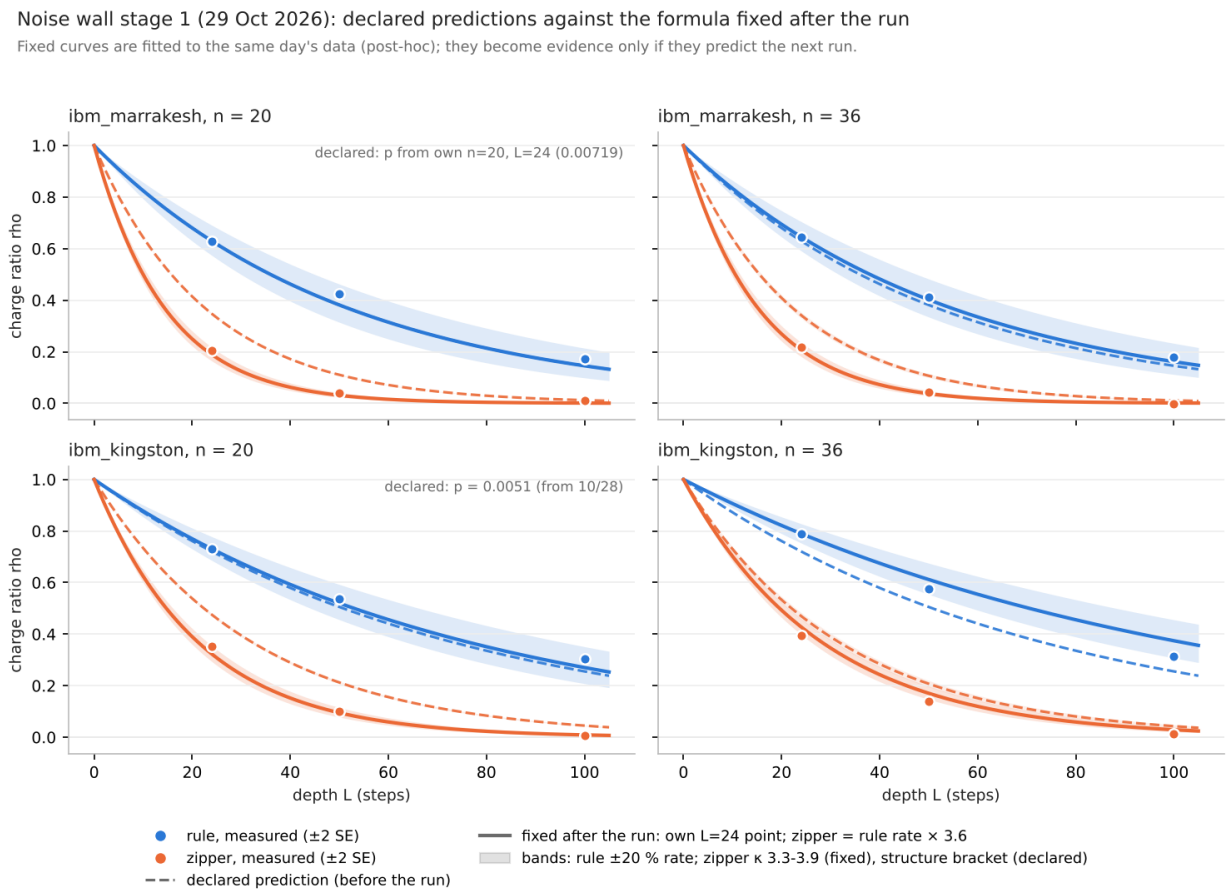


Figure 1: Declared predictions (dashed) against the formula fixed after the run (solid, with bands), for the rule (blue) and the zipper (orange) on both devices and both sizes. Points: measured ρ, ± 2 SE. The largest miss of the declared predictions is the zipper's: its structure ratio (about 2.3) underestimated its penalty (about 3.6) on all four panels. Fitted to the same day's data; script `readable_depth_figures.py`.

7.2 Declared tests and results

Test 2 ($n = 20, 36$) and part E of test 3 ($n = 40, 72$), $h_z = 0, 0.1, 0.3$, three placements, 500 shots, on `ibm_marrakesh` and `ibm_kingston`, 30 October 2026; declarations fixed before the fetch.

- **Rule A (who keeps more):** rule against each other arm at 2 SE, judged when either exceeds 2 SE. Over both devices: 71 wins, 11 ties, **0 losses**, 62 not judged. The lead does not depend on h_z.
- **Rule B (the fixed zipper penalty, $\kappa \in [3.3, 3.9]$ from Section 6):** BELOW wherever it could be judged ($\kappa = 1.7$–2.7, 18 cells); on marrakesh's part E the zipper sat at the noise floor in 5 of 6 cells (not judged), and the sixth, at 2.2 SE, came out ABOVE (5.4 ± 12). The penalty fixed on the charge did not transfer to the echo.

rule's L^* (P3, steps)	$n = 20$	$n = 36$	$n = 40$	$n = 72$
kingston, 12:22 CET	22.8–23.7	23.0–23.6	12.8–16.1	14.2–15.3
kingston, 14:12 CET (rule only, rerun)	21.9–22.3	24.5–26.2	13.0–16.0	14.6–16.0
marrakesh, 12:17 CET	14.6–14.8	14.9–15.8	14.1–15.3	14.3–15.5

Table 4: The rule's echo decay length over $h_z = 0$, 0.1, 0.3. On kingston it halves from $n = 36$ to $n = 40$; on marrakesh it does not depend on n. The rerun's declared reproduction test failed at 2 SE (12 of 21 cells agreed; differences of 0.02–0.05 in ρ_E), while the pattern over n and over sites repeated.

8 A defect the calibration does not show

Reading ρ_E per site instead of as a ring average explains the halving on kingston. Let $a_j = \frac{1}{4}\sum_{\text{inputs}}\langle -z_j Z_j\rangle$ be the raw echo signal of ring site j (1 perfect, 0 random, negative when the wrong bit is read more often than the right one). Figure 2 shows a_j at $D = 24$, $h_z = 0$ for three runs of the byte-identical $n = 40$ circuit (QPY SHA-256 `42376e3f...`) at 12:22, 14:12 and 15:24 CET, under three calibrations (11:55, 13:37 and 14:57 CET).

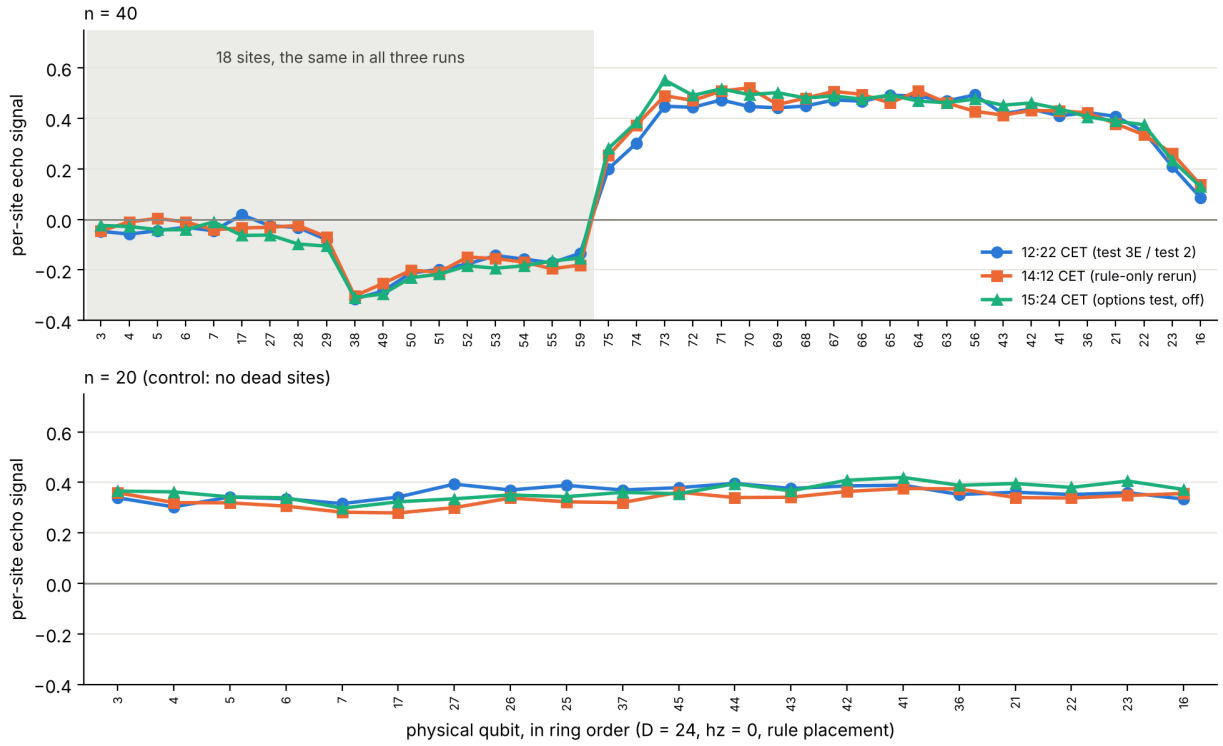


Figure 2: Per-site echo signal along the rule's ring, `ibm_kingston`, 30 October 2026, $D = 24$, $h_z = 0$. Top: $n = 40$, three runs of the same circuit; the shaded run of 18 sites is the same in all three. Bottom: the $n = 20$ ring, which uses qubits 3–7, 16, 17, 27 of the same region and shows no such run. Script `figures/defect_map_figure.py` on TOQB's records.

- **Reproducible and localized.** The same 18 sites (qubits 3–7, 17, 27–29, 38, 49–55, 59) fell below $a = 0.12$ in all three runs (site-wise correlation 0.98 between runs); their mean was

-0.133 ± 0.004 against 0.430 ± 0.003 for the other 22 (third run). Qubits 3–29 were near random, 38–59 inverted (-0.13 to -0.31). At $n = 72$ about 31 sites in two clusters behaved the same way in two runs; at $n = 20$ and 36, none.

- **Invisible to the calibration.** On these sites the day's CZ errors were 0.001–0.009, $T_2 \geq 125\,\mu s$ and readout errors $\leq 1.5\,\%$; each record's own readout reference was normal there.
- **Context-dependent.** Qubits 3–7, 16, 17, 27 are healthy on the $n = 20$ ring and qubits 27–29 on the $n = 36$ ring. The run appears when the ring passes 17–27–28–29–38. Which element or combination causes it is not known.

Two statements about these runs can be proved rather than estimated.

Lemma 1 (the circuit is not the cause). Pauli twirling of the CZ gates replaces each CZ by $P'\,\mathrm{CZ}\,P$ with P a random two-qubit Pauli and $P' = \mathrm{CZ}\,P\,\mathrm{CZ}$, again a Pauli because CZ is Clifford [12]. The ideal unitary, and so the ideal output (the flipped input), is unchanged. A change of the measured output under twirling is therefore a property of the device. Run with gate twirling (15:24 CET, otherwise identical), the inverted part moved from -0.214 to $+0.015$ and the near-random part from -0.052 to $+0.004$; the cause is on the device, and the inverted part is a coherent error that twirling turns into a random one. (The circuit was also checked noiselessly at $D = 2$ and 4, where every site returns the flipped input.)

Lemma 2 (how far one fault reaches). Let one fault E act after t of the echo's $2L$ steps, so that the noisy circuit is $U_2 E U_1$ with $U_2 U_1 = W$. Then $U_2 E U_1 = W\,(U_1^\dagger E U_1) = (U_2 E U_2^\dagger)\,W$. Each step moves support by at most one site each way along the ring (nearest-neighbour CZ ring and single-qubit gates; the rule's circuits contain no other two-qubit gates). So the effective error is supported within $\min(t, 2L - t) \leq L$ sites of E on either side, at most $2L + 1 = 25$ sites at $D = 24$, if errors act locally in space and time. The 18-site run at $n = 40$ fits one fault. At $n = 72$ the shortest arc of the ring that holds all dead sites is 43–45 positions long (two runs), which no single local fault can reach: there are at least two sources.

Options do not repair it. Dynamical decoupling (XY4) moved the 18 sites by $+0.018 \pm 0.005$ and cost the others -0.013 ± 0.004; gate twirling moved them by $+0.143 \pm 0.005$ (to about zero, not back to the healthy level) and cost the others -0.064 ± 0.004, and the healthy $n = 20$ ring a quarter of its signal (0.37 to 0.27). Neither was kept: the standard stays without both. That decoupling, which cancels static coupling on idle qubits, does almost nothing points away from idle crosstalk; this is a reading, not a test.

What the rest of the ring says. With the dead sites taken from one run and removed from the other (both ways, so no run is read on sites chosen from itself; post hoc), the remaining sites at $n = 40$ and 72 gave $\rho_E = 0.35$–0.45 at $D = 24$ and $L^* = 18.6$–21.3, close to the $n = 20$, 36 rings (22–26). Most of the drop with n in Table 4 sits in these runs. Masking recovers per-site quantities only; anything that needs every bit is lost once a dead site is in it (next section).

9 The global parity

$P = \prod_j X_j$ is conserved by the step at any n and has weight n: it is the widest quantity the ring has, and one bad site erases it. Declared (test 3, part P): the rule's volume V^* in $\rho_P = e^{-nL/V^*}$

inside a band around 171 (kingston) and 113 (marrakesh), and the zipper ratio in $[3.3, 3.9]$.

- **Failed:** the rule's V^* was 72.6 and 105.8 on kingston, 59.8 and 76.4 on marrakesh ($n = 20$, 36): 1 of 4 inside. The zipper's V^* was inside in 4 of 4; the ratio inside in 1 of 4 (1.7–2.6).
- **Duels at $n = 20$, 36:** rule 11 wins and 1 tie on kingston, 8 wins and 4 ties on marrakesh, no loss.
- **Above 36:** on marrakesh the readout reference (half-filled strings) fell to about 0.1 at $n = 40$ and to zero at $n = 72$; these rings include qubits whose readout errors were 17–34 % in the calibration: readout ends the parity there. On kingston the reference was 0.27 at $n = 40$, yet the parity was lost at $L = 1$ (-0.07 ± 0.06, against 0.54 at $n = 36$): the $n = 40$ ring is the echo's ring of Section 8. The longest exact ring of the day, $n = 88$ on kingston, gave 0.22 ± 0.10 at $L = 1$ (reported, not judged).

10 Output probe P2

[TODO: P2 tables; the contrast between a ring-averaged weight-2 charge and a single-site output]

11 Discussion

- **Selection and prediction.** The rule's schedules are fixed by formula; nothing is chosen as the best of several candidates by a calibration- or simulation-based score. For a selection of the best of K candidates by an estimate with error scale σ, the prediction is optimistic by up to $\sigma\sqrt{2 \ln K}$ [3]; the analogy with the Occam bound [4] is structural. This concerns the reliability of a prediction, not which placement keeps more signal, which we measured separately.
- **Why the duration matters.** Blind Qiskit placed the bare ring at equal gate count, yet its charge decayed 2–5 times faster per step than the rule's, with 5–9 times the duration. [TODO: idle decoherence, scheduling]
- **What a map buys.** D5 part A suggests about 0.3 in the classifier's slope at $n = 20$; between jobs, not declared. [TODO: a declared non-inferiority test with a stale map]
- **What a map cannot see.** The defect of Section 8 sat on qubits and couplers with good calibration numbers and appeared only in one combination of them. A placement that reads the map would not have avoided it; only running the circuit showed it. For the comparison of placements this cuts both ways: at $n \geq 40$ a placement's result also depends on whether its ring happens to pass such a site (the zipper's $n = 40$ ring on kingston was near random almost everywhere), so wins at $n \geq 40$ mix placement and the position of defects. The wins at $n = 20$, 36, where no such run was seen, do not.

12 Limitations

- Two days; one ring per size and placement. The defect was seen on one device and one day (three runs); whether it persists is open.
- The charge is a weight-2 quantity; L^* is not the depth at which an arbitrary output stays readable.

- The four-input ratio leaves a second-order relaxation bias.
- Every reference here is classically computable; no classical-hardness claim is made.

13 Reproducibility

TOQB [2]: `results/device-d6-2026-10-28`, `results/noise-wall-1-marrakesh-2026-10-29`, `results/noise-wall-1-kingston-2026-10-29`, `results/device-2026-10-30` (tests 2 and 3, the rule-only rerun with the post-hoc masking, the options test); declarations in `prereg/`. Circuit files over GitHub's size limit are kept off the repository with their SHA-256.

Acknowledgments

We acknowledge the use of IBM Quantum services for this work. The views expressed are those of the authors, and do not reflect the official policy or position of IBM or the IBM Quantum team. The device runs used the IBM Quantum Open Plan, including its one-time promotion of 180 minutes of runtime.

References

[1] TN Holdings, "Folding a Ring onto Heavy-Hex: Formula-Built Schedules with Provable Gate Counts, and What Their Signal Shows on Twins and Devices", version 1.2 (2026).

[2] TN Holdings, TOQB, the test bench for PSF-Zero's training and device runs (2026).

[3] J. E. Smith and R. L. Winkler, "The optimizer's curse: Skepticism and postdecision surprise in decision analysis", Management Science 52(3), 311–322 (2006).

[4] A. Blumer, A. Ehrenfeucht, D. Haussler and M. K. Warmuth, "Occam's razor", Information Processing Letters 24, 377–380 (1987).

[5] K. Maruyoshi et al., arXiv:2208.00576 (2023). [TODO: full reference]

[6] Oliva del Moral et al., arXiv:2603.13060 (2026). [TODO: full reference]

[7] D. C. McKay et al., "Benchmarking quantum processor performance at scale", arXiv:2311.05933 (2023). [TODO: check title]

[8] A. Pocklington and A. A. Clerk, arXiv:2604.18996 (2026). [TODO: full reference]

[9] R. E. Francois and G. R. Garrison, "Sound absorption based on ocean measurements. Part II: Boric acid contribution and equation for total absorption", J. Acoust. Soc. Am. 72(6), 1879–1890 (1982).

[10] R. J. Urick, *Principles of Underwater Sound*, 3rd ed., McGraw-Hill (1983).

[11] L. E. Fischer et al., arXiv:2411.00765 (2024). [TODO: full reference]

[12] J. J. Wallman and J. Emerson, "Noise tailoring for scalable quantum computation via randomized compiling", Phys. Rev. A 94, 052325 (2016).